

Week 2 Progress

From meeting recalibration to a manuscript-ready pathway panel

Stroke aggregates · seventeen labelled methods · real pollution

Bob Shen

Laidlaw Scholars Programme · Bishai Lab

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READY exposures · pollution · flu · 17-pathway dry-run

PENDING stroke aggregates · Gate 3 headline freeze

THIS WEEK one-command drop-in when files arrive

The story of Week 2 in one sentence

The meeting told us what the data cannot be.
The week built what the analysis must be.

17 July — hearing

- No admission reasons in the general HA file
- Near-term outcome: **stroke aggregates**
- Explore ~ 10 methods, not one hero model
- Stay complementary to daily heatwave–mortality work

This week — building

- Real EPD EPIC pollution (NO_2 , O_3 , $\text{PM}_{2.5}$, PM_{10})
- Pathway registry P01–P18 with SAP
- Dry-run: **17 pathways OK** (synthetic stroke)
- Drop-in command for real aggregates

Honesty line: synthetic coefficients prove plumbing — they are not findings.

What the lab meeting changed

Before	Meeting	Now
AMI + IS + HS principal-dx panel	Reasons for admission unavailable	Stroke aggregates; AMI out of scope from that file
One primary heat model	“Try ~10 methods”	Labelled pathway panel P01–P18
Heatwave metrics of interest	Align with Chao Ren / Wang EHWE	Spells, 2D3N, multi-def HW-month
Parallel to historical Goggins	Lab daily mortality HW work exists	Complementary estimand, not a duplicate

Insight: when the outcome is aggregate and the heatwave definition is contested, specification diversity is not noise — it is the scientific design.

Recalibrated question (Week 2 framing)

Among Hong Kong residents in the eligible aggregate population, how do **monthly** thermal exposures — continuous temperature, official extreme-day counts, Ren/Wang spell and 2D3N burden, and multi-definition heatwave-month indicators — associate with **stroke admission aggregates** (2013–2023), and how stable are those associations across a pre-labelled panel of specifications?

This estimand

- Monthly hospital *burden*
- Stroke aggregates
- Panel of labelled pathways

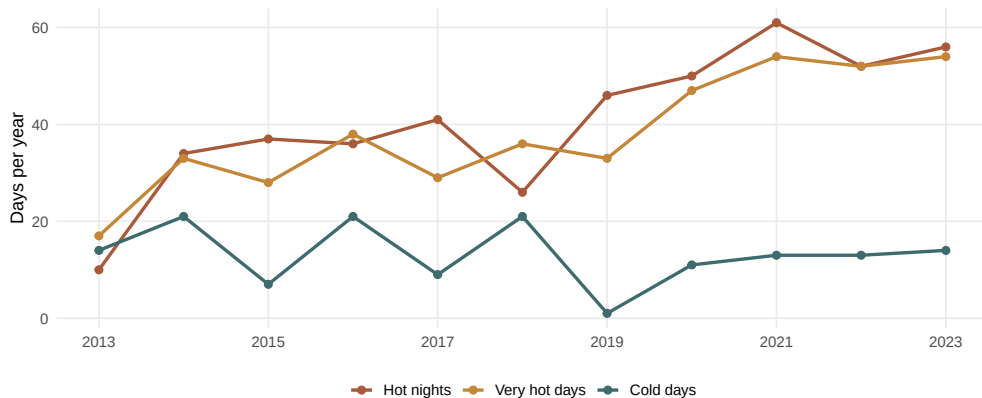
Not this estimand

- Daily DLNM triggering (Goggins)
- Excess-death heatwave counts (Liu/Ren/Bishai)
- AMI from files without reasons

Climate layer: still the project's ground truth

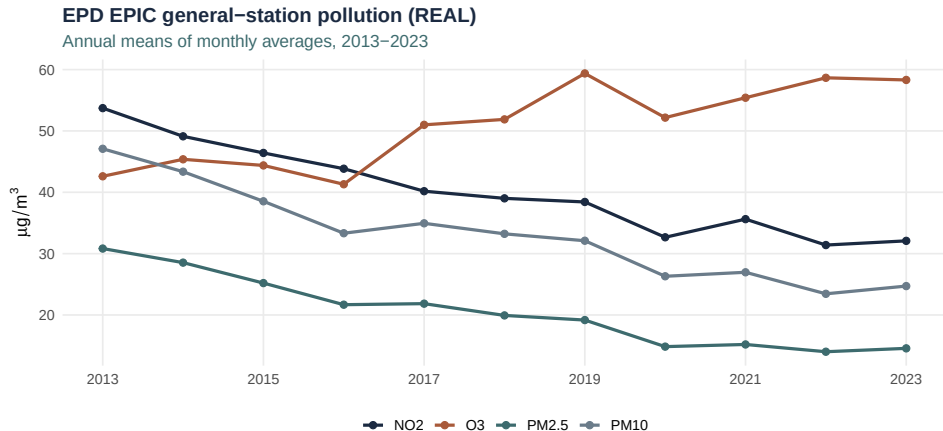
Official thermal extremes at HKO Headquarters

Annual counts, 2013–2023 (environmental data only)



Early period (2013–2018) averages ~31 hot nights / ~30 VHDs / ~16 cold days per year; late period (2019–2023) ~53 / ~48 / ~10. Peak hot-night year: **2021 (61)**. Cold days persist — they do not vanish.

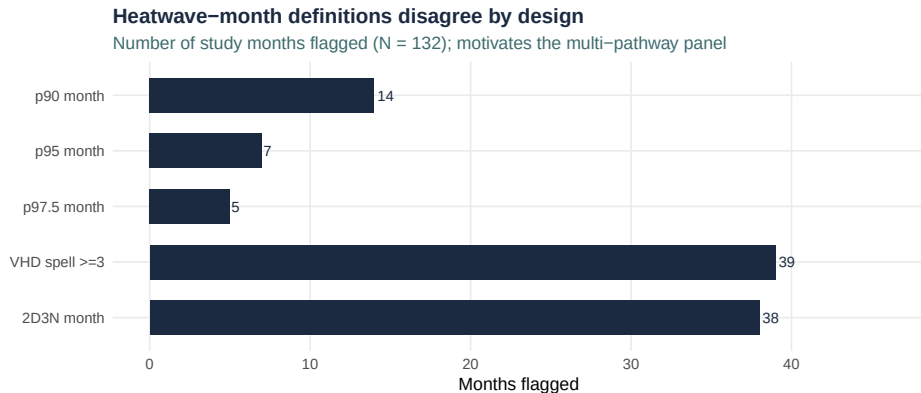
New this week: real pollution, not a placeholder



EPD EPIC monthly general-station means, 2013–2023.
Early (2013–2016) → late (2021–2023): NO₂ –32%, PM_{2.5} –45%, PM₁₀ –38%; O₃ +32%.

Modelling rule: stage pollution (none → NO₂ → PM_{2.5} → O₃ → multi). Enter O₃ last — heat-ozone collinearity.

Why definitions disagree — and why that is useful



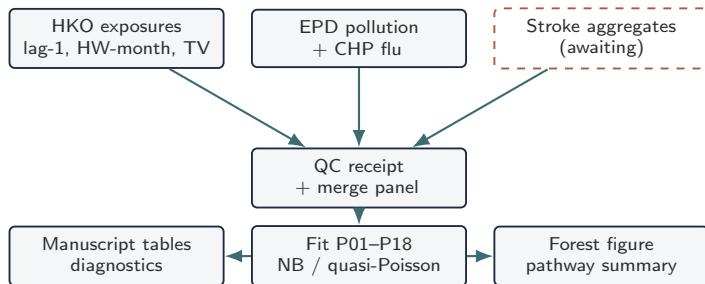
Guo et al. (EHP 2017) showed heatwave risk depends on percentile $\times$ duration grids. Wang/Ren (2019) add spell and combined day–night structure. Liu/Ren/Bishai mortality work treats multi-definition as first-class. **Our monthly collapse inherits that spirit** without claiming daily excess deaths.

The pathway panel: seventeen doors, one room

Family	Pathways
Continuous temperature / lag	P01 Tmean · P02 Tmax/Tmin (headline) · P03 lag-1
Extremes & heatwaves	P04 official (headline) · P05 spells · P06 2D3N · P07 HW-month
Cold	P08 cold-primary · P18 cold-month (p05)
Structure	P09 age · P17 sex
Confounding ladder	P10 COVID/holidays · P11 pollution · P12 AH · P14 flu
Sensitivity	P15 temperature variability · P16 pre-COVID
Optional	P13 IS vs HS — <i>enable only if subtype exists</i>

Reporting rule: one panel, not seventeen discoveries. Gate 3 freezes the headline pair — proposal **P02 + P04** — after real descriptives.

Architecture: what runs when the file arrives



`PATHWAY_MODE=real Rscript scripts/run_pathway_pipeline.R`

Real mode refuses synthetic outcomes. Place approved CSV under `data_raw/ha_secure_placeholder/` — never commit microdata.

Dry-run proof: plumbing works

What succeeded

- 17 / 17 enabled pathways returned ok
- P13 correctly disabled (no subtype)
- Age×sex synthetic grain exercised P09/P17
- Flu pathway used complete-case (121/132 months)
- Forest + diagnostics + manuscript tables written

What it does not mean

- Coefficients are **not** findings
- Synthetic stroke is a schema rehearsal
- Gate 1–2 still own real data arrival
- Gate 3 still owns the headline freeze

Week 1 parallel: then, HKO descriptives were real and models were synthetic. Now, *all public layers* are real — only the outcome remains the missing keystone.

Readiness: one missing piece

Week 2 readiness: one missing piece

Public layers and analysis plumbing are ready; stroke aggregates unlock inference



Literature that shaped the panel (selected)

- **Goggins 2012/2013** — cold-dominant stroke/AMI; subtype and sex matter → P08, P13, P17
- **Guo 2024** — official hot-night binary often null; intensity matters → P04 necessary, P05/P06 essential
- **Wang/Ren 2019** — VHD/HN spells and 2D3N → P05, P06
- **Guo/Gasparrini 2017** — percentile $\times$ duration grids → P07 multi-def HW-month
- **Yang/Chong 2025** — cold + influenza and stroke (~ 1.17 M admissions) → P14, P18
- **Tian/Qiu 2023 (JoGH)** — temperature variability and circulatory admissions → P15
- **Guo 2025** — pollution amplifies temperature–hospitalisation associations → P11 staged

Full map: `literature/pathway_evidence_memo.md`

Scientist operating system (new documents)

Contracts

- Statistical Analysis Protocol
- Pathway catalogue + YAML registry
- Stroke aggregate JSON schema
- Decision gates (updated for P01–P18)

Products

- Pathway coefficient panel
- Headline residual diagnostics
- Forest figure
- Manuscript table shells
- Scientist runbook (drop-in steps)

Design ethic: every estimate carries a `pathway_id`. Amendments after Gate 3 require a new pathway ID — no silent redefinition of P02/P04.

Still open — decisions that unlock inference

Open item	Working default	Owner
Stroke aggregate file + dictionary	Schema: <code>month_id</code> , <code>n_events</code> minimum	Roro / Bob
Grain (territory vs age×sex)	Prefer age×sex if available	On file inspection
Subtype IS/HS	Enable P13 only if present	On file inspection
Governance / use path	Follow PI determination	Bishai
Headline pair (Gate 3)	Proposal: P02 + P04	Team after descriptives
Official holiday gazette	Deterministic scaffold v2 until then	Bob (optional)

Ask: when aggregates are cleared for transfer, drop the CSV and run real mode. First deliverable within hours: QC receipt + pathway panel + forest — then Gate 3 freeze.

Week 2 close

We no longer wait to invent the analysis.

We wait only for the file.

- Public layers: HKO, C&SD, EPD, CHP flu — **real**
- Analysis system: 17 pathways, SAP, diagnostics — **ready**
- Outcome: stroke aggregates — **pending**
- Claim discipline: monthly burden $\neq$ daily DLNM $\neq$ excess deaths

Repo: <https://github.com/bobshenruililin/Laidlaw-Heat-Project>

Companion brief: [reports/Week2_Progress_Brief.pdf](#)

Appendix

Numbers, command, and pathway index

A1 — Verified environmental numbers (Week 2)

- Study months: $N = 132$ (Jan 2013–Dec 2023)
- Mean temperature: 24.02°C (SD from monthly series)
- Lag-1 T_{max} complete for all 132 months (Dec 2012 joined)
- Heatwave-month flags: $p_{90} = 14$, $p_{95} = 7$, $p_{97.5} = 5$; 2D3N months = 38
- Temperature variability: mean daily diurnal range $\approx 4.56^{\circ}\text{C}$
- Flu Express coverage: 121/132 months (early-2013 gap documented)
- Pollution provenance: EPD_GENERAL_STATIONS (placeholder removed)

A2 — One-command drop-in

```
# Dry-run (synthetic stroke; current default)
```

```
Rscript scripts/run_pathway_pipeline.R
```

```
# Real aggregates
```

```
# 1. Place approved CSV in data_raw/ha_secure_placeholder/
```

```
# 2. Then:
```

```
PATHWAY_MODE=real Rscript scripts/run_pathway_pipeline.R
```

Outputs: outputs/reports/pathway_panel_summary.md, outputs/tables/pathway_panel_estimates.csv,
outputs/figures/pathway/pathway_panel_forest.png.

A3 — Pathway index (enabled unless noted)

ID	Status	Title
P01	ok	Continuous same-month mean temperature
P02	ok	Continuous same-month Tmax and Tmin
P03	ok	Lag-1 month temperature
P04	ok	Official extreme-day counts
P05	ok	Ren/Wang spell burden
P06	ok	Combined day–night 2D3N burden
P07	ok	Heatwave-month multi-percentile + spells
P08	ok	Cold-side primary
P09	ok	Age-stratified (65+ vs younger)
P10	ok	COVID and holiday sensitivity
P11	ok	Pollution-staged on P02
P12	ok	Absolute humidity sensitivity
P13	disabled	Subtype IS vs HS (await fields)
P14–P18	ok	Flu, TV, pre-COVID, sex, cold-month